

MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

SANTOSH, TANGAIL-1902



DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report

Lab Report No : 03
Lab Report Name : Cross Cable Configuration using CAT6 cable
Course Title : Computer Networks Lab
Course Code : ICT-3202

Submitted By	Submitted To
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Date of Performance :

Date of Submission :

Experiment No: 03

Experiment Name: Cross Cable Configuration Using CAT6 Cable

Objectives:

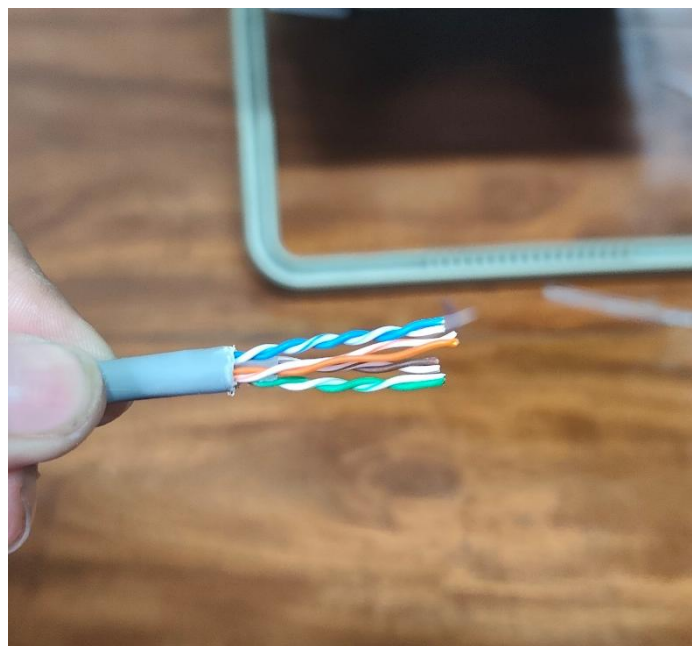
1. To understand the concept and structure of a CAT6 cable.
2. To learn the standard wiring color sequence used for cross-through cable connections.
3. To practically prepare a cross cable using CAT6 cable, RJ45 connectors, and a crimping tool.
4. To test the connectivity of the prepared cross cable using a cable tester.
5. To understand the practical application of cross cables in networking (connecting similar devices, e.g., PC to PC, Switch to Switch).

Necessary Materials and Software:

1. CAT6 Cable (UTP)
2. RJ45 Connectors (2 pieces)
3. RJ45 Crimping Tool
4. Cable Cutter/Stripper
5. Cable Tester (LAN Tester)

Procedure:

1. Cut the Cat6 cable to the desired length using a cable cutter.
2. Using the stripper on the crimping tool, remove about 1 inch of the outer jacket from both ends of the cable, exposing the four twisted pairs (8 wires).



3. Untwist the pairs and arrange the wires on One End according to the T568B standard:

Pin 1: White-Orange

Pin 2: Orange

Pin 3: White-Green

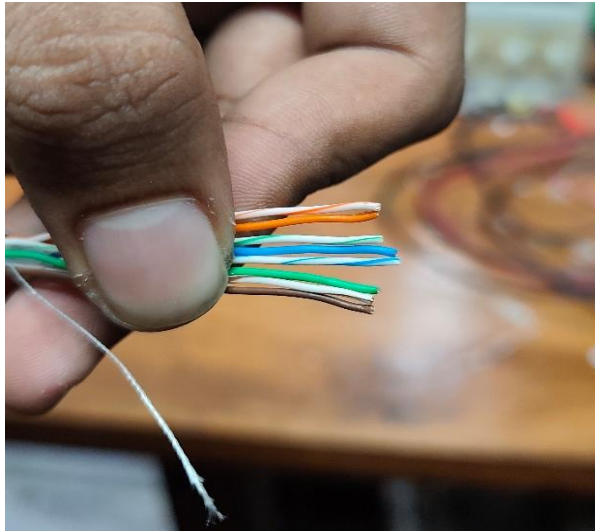
Pin 4: Blue

Pin 5: White-Blue

Pin 6: Green

Pin 7: White-Brown

Pin 8: Brown



4. Arrange the wires on the Other End according to the T568A standard:

Pin 1: White-Green

Pin 2: Green

Pin 3: White-Orange

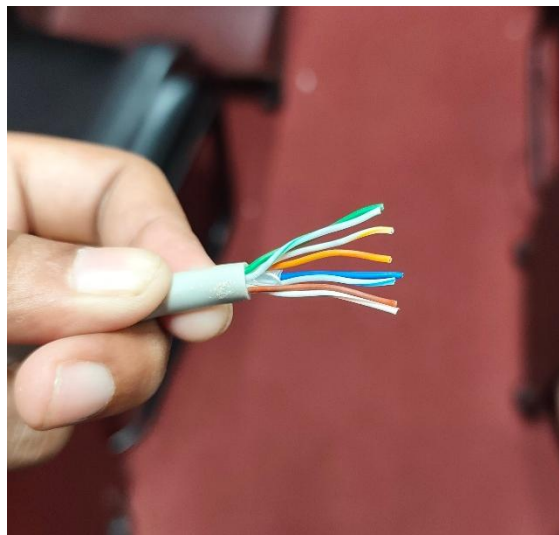
Pin 4: Blue

Pin 5: White-Blue

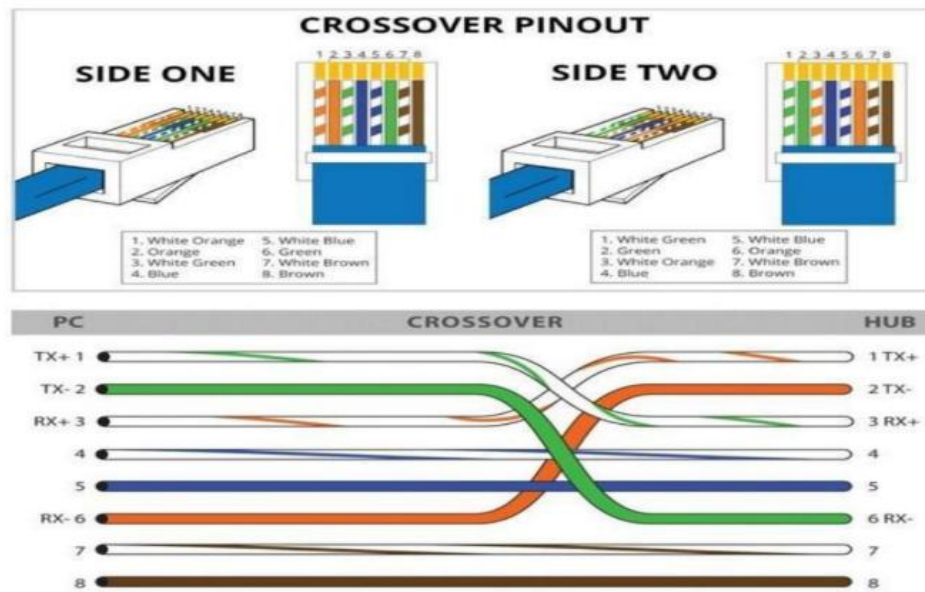
Pin 6: Orange

Pin 7: White-Brown

Pin 8: Brown

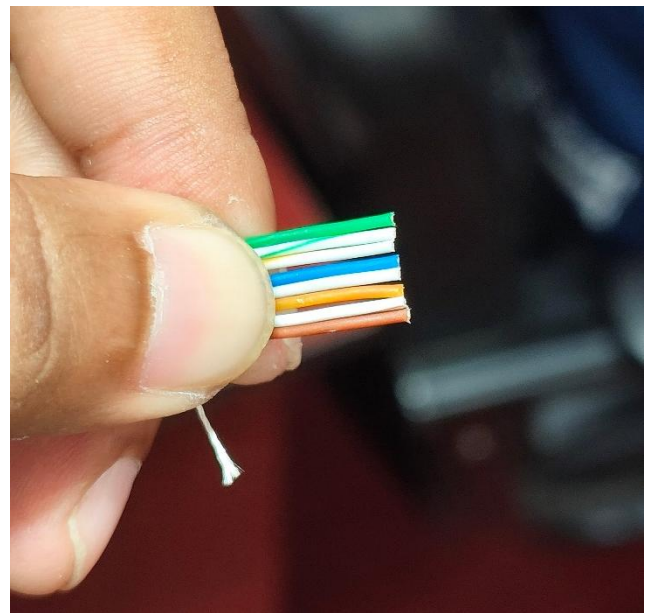


Crossover Cable Configuration

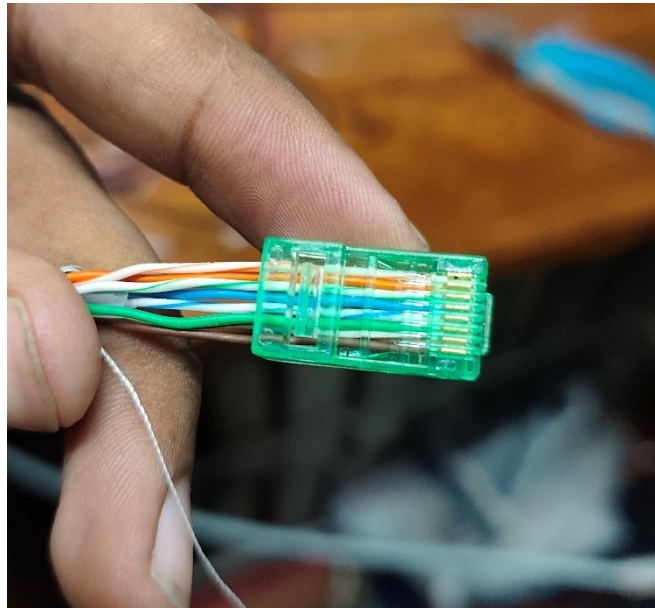


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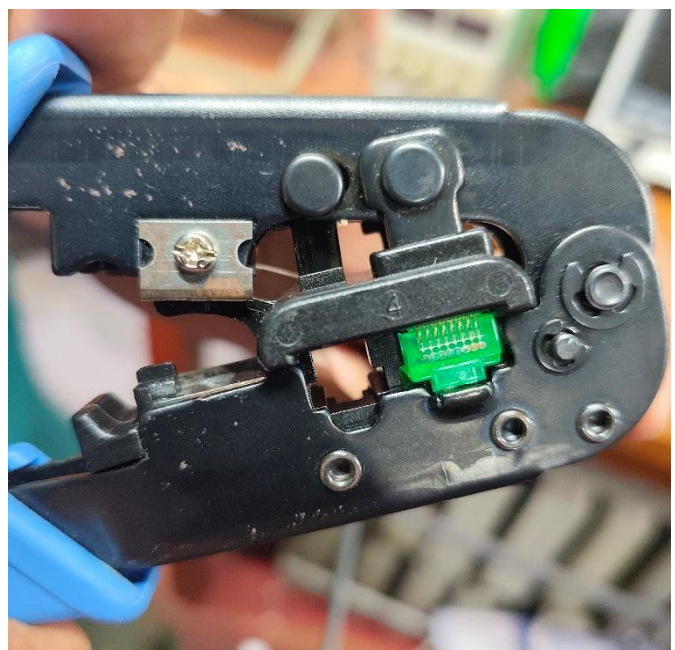
5. Straighten and flatten the wires, then trim them evenly to about 0.5 inch so they fit properly into the RJ45 connector.



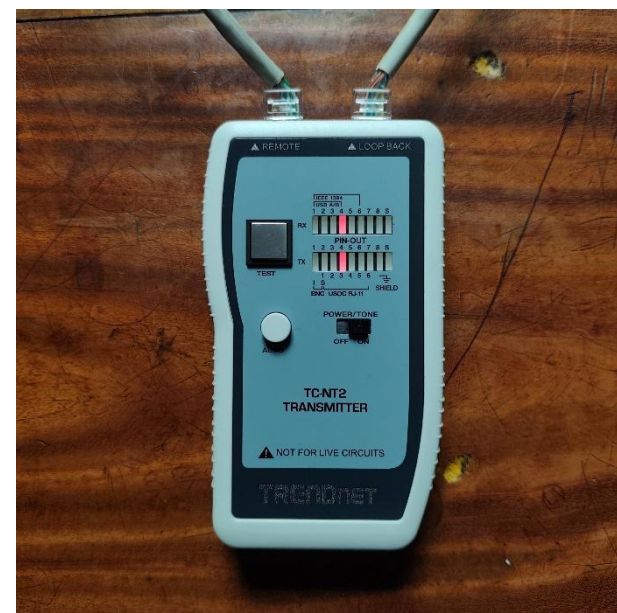
6. Insert the wires into the RJ45 connector in the correct order, making sure each wire reaches the front end of the connector and the outer jacket is inserted slightly into the connector for strain relief.

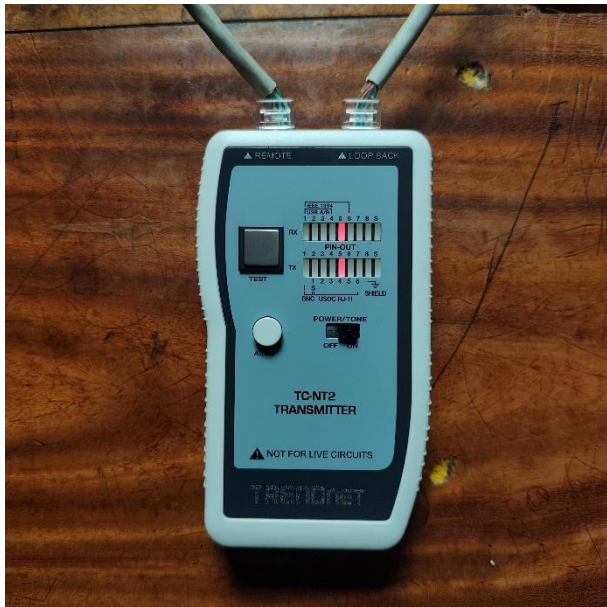


7. Place each connector into the crimping tool and press firmly to crimp the connector, securing the copper pins onto the wires.



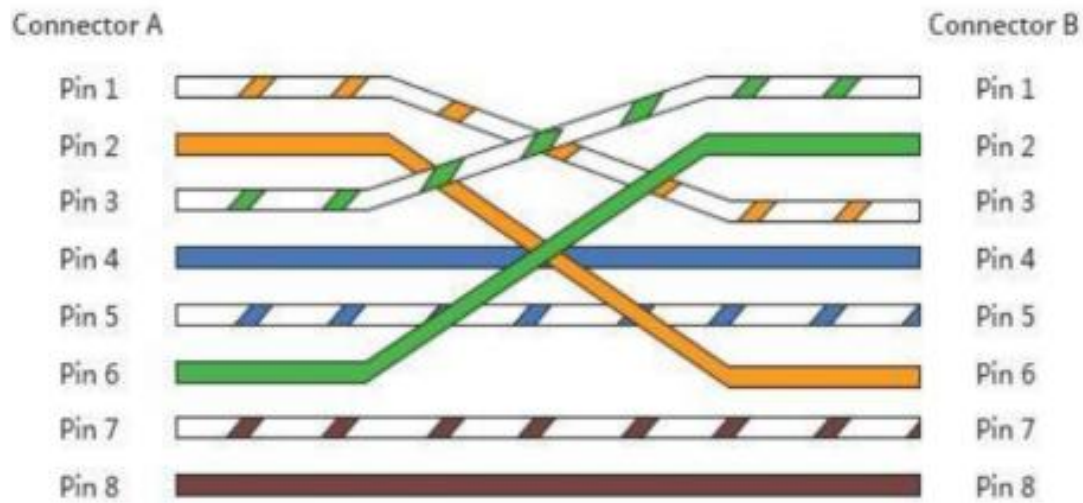
8. Insert both ends of the finished cable into a cable tester to check connectivity.





9. Since it is a cross cable, the tester LEDs will show a crossed pattern:
 Pin 1 to Pin 3
 Pin 2 to Pin 6
 Pin 3 to Pin 1
 Pin 4 to Pin 4
 Pin 5 to Pin 5
 Pin 6 to Pin 2
 Pin 7 to Pin 7
 Pin 8 to Pin 8

Crossover Cable Wiring Scheme



Conclusion:

In this experiment, a crossover Cat6 cable was successfully prepared by wiring one end using the T568B standard and the other end using the T568A standard. The cable was tested using a cable tester, confirming the correct crossed pin connectivity (transmit and receive pairs swapped). This type of cable is used to directly connect similar networking devices, such as two PCs or two switches, without requiring an intermediate device. Through this experiment, practical understanding of the differences between straight and cross cable wiring standards, along with hands-on cable termination and testing skills, was achieved.